

TRYVANNSHOGDA, Norway

WMO: 014900

Lat: 59.984N

Lon: 10.669E

Elev: 1732

StdP: 13.80

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	9.1	1.1	6.2	9.2	4.4	7.4	11.9	23.8	27.7	21.3	26.9	4.4	50	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	74.6	59.5	71.4	58.5	68.0	57.0	62.1	69.6	60.4	67.2	58.9	65.3	6.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.5	80.8	63.6	58.0	76.5	62.2	56.3	72.1	61.0	28.7	69.7	27.5	67.1	26.5	65.5	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	17.0	14.9	4.5	78.3	4.4	5.1	1.4	81.9	-1.2	84.9	-3.7	87.8	-6.9	91.5
			4.0	63.7	4.4	2.0	0.8	65.1	-1.7	66.3	-4.2	67.5	-7.4	68.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	24.6	25.1	30.7	39.1	47.6	54.6	59.1	56.9	49.6	39.7	32.3	27.6
	DBStd	13.84	7.63	7.77	7.82	6.43	7.58	5.76	5.55	4.71	4.89	5.94	7.01	7.93
	HDD50	4184	787	696	598	333	140	15	0	3	66	321	530	695
	HDD65	8914	1252	1116	1063	777	541	319	202	256	463	785	980	1160
	CDD50	776	0	0	0	6	66	153	282	215	53	1	0	0
	CDD65	32	0	0	0	0	2	7	19	4	0	0	0	0
	CDH74	111	0	0	0	0	5	23	79	4	0	0	0	0
	CDH80	6	0	0	0	0	0	0	6	0	0	0	0	0
Wind	WSAvg	7.1	7.1	6.6	7.2	7.1	7.1	6.7	6.5	6.5	7.4	7.7	7.6	7.6
Precipitation	PrecAvg	37.7	3.0	2.3	2.1	2.4	2.5	3.2	3.5	4.3	3.5	4.2	4.0	2.8
	PrecMax	53.4	7.4	7.2	6.1	4.6	5.6	6.1	7.1	7.9	8.3	10.1	14.8	5.8
	PrecMin	29.2	0.6	0.7	0.0	0.2	0.1	0.3	0.6	0.9	0.8	0.4	0.9	0.6
	PrecStd	5.8	1.7	1.4	1.3	1.2	1.4	1.5	1.6	1.7	1.7	2.4	2.8	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	44.8	53.9	60.4	73.7	77.3	80.1	74.0	65.8	54.5	49.2	44.5
		MCWB	37.5	37.1	43.8	47.5	56.2	58.7	62.0	60.3	56.1	49.4	46.0	41.2
	2%	DB	39.1	41.0	48.6	56.6	68.2	72.7	76.1	70.9	62.0	51.6	46.1	41.8
		MCWB	36.2	35.1	39.9	45.8	54.0	56.7	60.8	59.8	53.7	48.6	44.3	39.2
	5%	DB	36.6	37.7	44.4	52.7	64.1	68.4	72.8	67.6	59.1	49.9	43.7	39.6
		MCWB	34.2	33.5	37.2	43.5	51.8	54.9	60.2	58.3	52.5	47.7	42.3	37.5
	10%	DB	34.3	34.8	40.9	49.2	59.9	64.4	69.0	64.5	56.5	47.9	41.5	37.5
		MCWB	32.5	32.6	34.9	41.2	50.3	53.0	58.6	56.8	51.5	45.6	40.2	35.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	39.3	44.8	50.1	59.3	60.8	64.8	63.5	57.9	52.5	47.6	42.4
		MCDB	40.6	43.0	52.5	57.9	67.2	70.5	74.8	69.5	62.1	53.1	48.0	43.1
	2%	WB	36.7	36.4	40.4	47.0	56.4	58.6	63.0	61.5	55.8	50.1	44.8	40.1
		MCDB	38.6	39.2	47.2	54.7	64.7	67.8	72.1	66.6	59.3	50.6	45.7	41.2
	5%	WB	34.6	34.4	38.0	44.4	53.7	56.7	61.2	60.0	54.1	48.3	42.5	38.1
		MCDB	36.2	36.6	43.1	50.9	61.4	65.4	69.3	64.8	56.9	49.4	43.2	39.2
	10%	WB	32.5	32.7	35.8	42.2	51.1	54.8	59.7	58.4	52.8	46.3	40.4	35.7
		MCDB	33.9	34.4	39.5	48.2	58.2	62.3	67.0	62.9	55.5	47.5	41.4	37.0
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.1	9.4	11.4	12.7	13.4	12.4	11.0	10.0	8.0	6.4	6.6
		MCDBR	8.6	10.4	13.7	15.5	16.7	17.3	17.0	13.8	13.3	9.3	8.7	8.6
		MCWBR	7.5	7.7	8.0	8.5	8.6	8.8	7.5	6.4	7.8	7.3	7.8	8.2
	5% WB	MCDBR	7.9	8.5	12.4	14.3	14.5	14.6	15.2	11.5	10.7	7.9	8.6	8.6
		MCWBR	7.2	6.9	8.1	8.5	8.4	8.5	7.5	6.6	7.6	7.5	8.3	8.5
Clear-Sky Solar Irradiance	taub	0.240	0.280	0.305	0.343	0.341	0.332	0.350	0.344	0.315	0.303	0.257	0.211	
	taud	2.227	2.333	2.385	2.388	2.458	2.498	2.459	2.472	2.545	2.502	2.307	2.118	
	Ebn at Noon	180	234	265	272	282	286	277	270	261	226	171	141	
	Edh at Noon	12	20	26	31	31	30	31	29	23	17	11	9	
All-Sky Solar Radiation	RadAvg	90	284	753	1194	1549	1773	1584	1246	826	365	121	60	
	RadStd	17	52	81	161	159	133	205	100	95	71	33	6	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (58 neighbors)	N/A	N/A	+2.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page